Refinery Operations Field Handbook & Operating Envelopes

Document ID:	APEL-MAN-OPS-101	**Effective Date:**	2026-08-01
Revision:	5.0	**Classification:**	CONFIDENTIAL - STRICT PROPRIETARY
Department:	Operations Department	**Site Location:**	Jamnagar Refinery Complex (JRC)

1. Refining Philosophy

ApexPetro Jamnagar Refinery operates 34 MMBPA integrated processing capacity. Primary conversion units (CDU-4, HCU-1) must operate within designated Safe Operating Envelopes (SOE).

2. Unit Feed and Temperature Envelopes

- Crude Distillation Unit 4: Feed throughput 120 - 150 kbd; furnace transfer line temp 355 - 365 °C.
- Hydrocracker Unit 1: Reactor pressure 140 - 148 bar(g); recycle gas purity minimum 96.5% H2.
- Dispatch Terminal DISP-1: Diesel export temperature maximum 45 °C; pipeline pressure 12 - 18 bar(g).

3. Field Operator Walkaround Rounds

Console and field operators must conduct physical battery limit inspections every 2 hours, logging bearing temperatures, gland leakage rates, and local pressure gauges.